

BEC303 Network Analysis — Assignment 2 — Applying and bounding the network theorems

Module 2: Network Theorems for DC and AC Circuits

Field	Value
Maximum marks	20
Weightage	10% of CIE
Issue date	< <TBD> >
Due date	< <TBD> >
Items	10 problems + 3 theory questions
CO / Bloom coverage	CO2: 13 · L1–L2 0, L3–L4 9, L5–L6 4

Objective

Develop judgement about which theorem to apply and, more importantly, where each one stops being valid — with particular attention to power, dependent sources and the difference between maximum power and maximum efficiency.

Learning outcomes

- Student will be able to apply superposition correctly, including across sources at different frequencies.
- Student will be able to construct Thévenin and Norton equivalents for networks with and without dependent sources.
- Student will be able to distinguish reciprocal from non-reciprocal networks and justify the classification.
- Student will be able to select a matching criterion appropriate to a stated design constraint.

Part A — Problems (17 marks)

P1. For a two-source resistive network of your construction, find a chosen branch current by superposition, then verify the same result by nodal analysis. Report the contribution of each source as a percentage of the total. [C02 | L3 | 2 marks]

P2. Using the P1 network, compute the power in one resistor by the incorrect route (superposing partial powers) and by the correct route. Quantify the error and identify the cross term explicitly. [C02 | L4 | 2 marks]

P3. Solve a series RC branch driven simultaneously by a DC source and a sinusoidal source, and present the capacitor voltage as an explicit function of time. [C02 | L4 | 2 marks]

P4. Find the Thévenin equivalent of a network containing only independent sources by the look-back method, and verify it with both the open-circuit and the short-circuit limiting cases. [C02 | L3 | 2 marks]

- P5.** Find R_{th} for a network containing a dependent source using the 1 V test-source method. Then compute what the answer would have been under the illegal deactivation shortcut and state the percentage error. [C02 | L4 | 2 marks]
- P6.** Convert your P4 Thévenin equivalent to Norton form and demonstrate that both reproduce the same load voltage for three different load resistances. [C02 | L3 | 2 marks]
- P7.** Verify the reciprocity theorem numerically on a passive ladder of at least four resistors, then add a dependent source to the same ladder and show that reciprocity now fails. [C02 | L4 | 2 marks]
- P8.** For a source with $V_{th} = 24$ V and $R_{th} = 75\ \Omega$, plot delivered power against load resistance from $10\ \Omega$ to $750\ \Omega$, mark the maximum, and annotate the loads that deliver 90 % of it. [C02 | L4 | 1 marks]
- P9.** For an AC source with $Z_{th} = 40 + j60\ \Omega$, find the conjugate-matched load, the maximum delivered power, and the power that would be delivered by a purely resistive $72\ \Omega$ load instead. [C02 | L4 | 1 marks]
- P10.** Take one real design — an RF receiver front end, an audio output stage, or a sensor amplifier — and state which matching criterion applies, why, and what would go wrong under each of the other two criteria. [C02 | L5 | 1 marks]

Part B — Theory (3 marks)

- T1.** Explain why superposition and Thévenin remain valid in the presence of dependent sources while reciprocity does not. Your answer must identify the specific hypothesis that dependent sources violate. [C02 | L5 | 1 marks]
- T2.** Argue that maximum power transfer and maximum efficiency are incompatible design goals, and give one engineering context in which each is the correct choice. [C02 | L5 | 1 marks]
- T3.** Explain why two limiting-case checks, rather than one, are needed to verify a Thévenin equivalent, and construct a wrong equivalent that passes the open-circuit check but fails the short-circuit check. [C02 | L5 | 1 marks]

Submission guidelines

- Format: PDF, single file
- File name: <USN>_BEC303_Assignment2.pdf
- Channel: LMS upload
- Late policy: 10% per day, nothing after 5 days
- Every numerical answer must carry units and a stated reference direction.

Assessment rubric

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Correct application of theorems	7	Every theorem applied with its hypotheses checked and stated; all numerical results correct and verified.	Theorems applied correctly; hypotheses implied rather than stated; minor arithmetic slips.	Theorems applied mechanically with no hypothesis check; several numerical errors.	Theorems misapplied, e.g. dependent sources deactivated or superposition used for power.

Criterion	Weight	Excellent	Good	Satisfactory	Needs improvement
Treatment of validity limits	5	Every limit identified and quantified, including the power cross term and the dependent-source error, with percentages given.	Limits identified and described qualitatively.	Limits mentioned but not demonstrated.	Limits absent; incorrect results presented as valid.
Verification and cross-checking	4	Every major result verified by an independent route or by both limiting cases.	Most results verified by at least one check.	One or two checks performed.	No verification attempted.
Engineering judgement in the design question	4	Matching criterion chosen with an explicit scarce-resource argument and the failure mode of each alternative described.	Criterion chosen with a sound reason; alternatives briefly addressed.	Criterion stated without a supporting argument.	Maximum power transfer assumed to be universally desirable.
Total	20				

Course outcome attainment

Course outcome	Marks carried
CO2	20
Total	20

Academic integrity

Individual submission. Identical submissions score zero for both parties.